

Nothing to See Here: How to Say It When You Need to

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Now is a great time to commit financial fraud. We show decades-long, strikingly suspicious return patterns in the stocks of many of the world's publicly traded companies as examples of what you can get away with if you try. We show the willingness of regulators, government officials, academics, journalists, and others to turn a blind eye to evidence of malfeasance with a publicly maintained, years-long email thread with the world's financial regulators. We use articles from the *Financial Times* to provide specific, practical examples of how to say there is nothing to see here when you need to. We analyze the incentive structure created by a type of market manipulation available to large quantitative hedge funds to provide specific, practical suggestions for how to design your fraud to maximize your profit and minimize your risk. These suggestions, honed and tested for years in one of the largest financial frauds in history, should serve as useful and practical guidance to you in running your own financial fraud.

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I. NOTHING TO SEE HERE

Perhaps you are reluctant to start your own financial fraud because you fear getting caught. To put your mind at ease and demonstrate the power of the techniques in this article, we start by showing reassuring examples from around the world of the blatantly incriminating evidence you can produce and get away with.

Figure 1 explains the format of the plots in this article. Our previous articles discussing overnight and intraday returns [1–6] mostly used a linear vertical scale, as in Figure 1(a). All plots in this article use the format of Figure 1(c), which shows the informative log curve in Figure 1(b) and the intuitive numbers in Figure 1(a).

Figure 2 shows what you should expect the remaining figures to look like.

Figure 3 shows overnight and intraday returns to many of the world's major stock market indices.

Figure 4 shows overnight and intraday returns to 50 stocks included in a public thread with the United States Securities and Exchange Commission (SEC) and others [7]. This public thread is discussed further in Section III.

Figures 5, 6, and 7 show overnight and intraday returns to 25 stocks selected from France's CAC 40 index, Germany's DAX index, and the United Kingdom's FTSE 100 index, respectively.

Figure 8 shows overnight and intraday returns to 25 stocks in India's Nifty 50 index. We recommend making a big print of Figure 8 and hanging it on your wall.

Whenever you get worried about your fraud being discovered, look at your wall and be reassured.

Figures 9 and 10 show overnight and intraday returns to 25 stocks in Japan's Nikkei 225 index and China's SSE 50 index, respectively.

As in our previous articles on this topic [1–6], you can reproduce any plots you wish yourself. We use data publicly available from Yahoo! Finance [8]. Our code is publicly available at Ref. [9].

We show six full pages of plots – one page of setup (Figures 1–2) followed by five pages with a total of 224 plots from around the world spanning decades (Figures 3–10) – to demonstrate the extraordinary power of the techniques in this article. If your fraud is properly designed (Section III), you can leave footprints as blatant as Figures 3–10 around the world for decades. Few people will notice, those few will keep it to themselves, and nobody will raise a fuss.

This important point holds even if the cause of Figures 3–10 turns out to be innocuous. Few people noticed Figures 3–10, those that did don't have a good reason to believe Figures 3–10 are fine [5, 7], they kept it to themselves, and none of them raised a fuss.

Figures 3–10 and the global, decades-long absence of a fuss should give you the confidence you need to follow our guidance and launch your own financial fraud.

If anyone does raise a fuss, you can dismiss even evidence as blatantly incriminating as Figures 3–10 with brilliant, well-tested techniques used by the *Financial Times* to say there is nothing to see here [35].

II. HOW TO SAY IT

If your fraud is well designed, you yourself won't have to say there is nothing to see here. You can get somebody else to say it for you. Watching people stumble over themselves in their attempts to say there is nothing to see here on your behalf is very fun.

A January 2022 *Financial Times* column [10] provides a brilliant collection of techniques for how to say there is nothing to see here. The column is freely available at

Ref. [10], and we highly recommend you pause here and read it in full before continuing. See how many of the following clever, skilled, artful techniques you can spot. If you ever need to say there is nothing to see here, you or your mouthpiece [36] can use these practical, effective, well-tested techniques to make people and problems go away.

In the face of lots of evidence, focus on the least damning. Faced with all the plots in our previous articles [1–6], just show the top left plot of Figure 3.

Understate the magnitude of the evidence with numbers. If throwing dividends into the trash makes the problem look smaller, then throw dividends into the trash.

Understate the magnitude of the evidence with plots. After throwing dividends into the trash to reduce the height of the overnight curve, make the intraday curve look better by adding 100%. Cumulative returns of course start at 0%, not 100%, but if the cumulative intraday return ends near 0% and you don't want it to, just add 100% to everything so it looks like the cumulative intraday curve ends near 100%. Do not explicitly provide the final cumulative returns at the right of the plot (like we do in all our plots [1–6]), since this makes your subterfuge too obvious and unnecessarily gives people concrete numbers to fuss about.

Understate the magnitude of the evidence with words. Euphemistically describe negative returns of -84% and worse (including the intraday returns in Japan and Norway in Figure 3) as “gentle intraday declines” [37].

Having understated the evidence, stick with it. Don't give an inch. If somebody asks you to remake your plot without throwing dividends into the trash [11], refuse. Repeatedly show the same plot throwing dividends into the trash again [12] and again [13].

Frame the evidence to your advantage, focusing on aspects of the evidence that seem easier to explain and ignoring evidence that seems harder to explain. Frame the evidence as “stocks in the night-time” [10] or “overnight drift” [13]. Focus on stocks going up overnight. Don't focus on stocks going down intraday because you can't explain that in terms of returns being due to the bearing of risk. Don't focus on the striking, decades-long consistency of many of the plots in Figure 3 and our previous articles [1–6] because you can't explain that in terms of the uncoordinated actions of many traders. Don't even mention the many stocks (including BCS, EFX, and HPQ in Figure 4) in which the pattern is flipped.

Reference articles that don't say what you claim they say. Say the fraud you are accused of is impossible because your trading costs would be too high, referencing a blog post discussing the trading costs of a strategy different from the one you are accused of.

Point out superficially related, less compelling facts that are unrelated to you (or at least hard to tie to you). Point to something in a different market. Faced with damning evidence in the stock market, talk about the futures market.

Suggest many other explanations for the evidence. None of them need to be compelling. If you offer only one alternative explanation, you may have to defend it. If you offer many alternative explanations, you won't have to defend any of them. Even better, suggest the explanation is “a mix of” [13] your many explanations.

Offer vague, poorly specified explanations. Each of your alternative explanations should have few words. Don't commit to an explanation in an academic paper, which may be specified in sufficient detail to be proven wrong. Just casually riff whatever comes to mind. A vague explanation makes skeptics do all the work trying to fill in the details and gives you plenty of wiggle room to squirm out of whatever nonsense you might say. “Index funds buying in the closing auction” [10] is in the wrong direction, but you can smoothly turn the short phrase into “the rising influence of index funds” [13] without missing a beat.

Offer a vague, poorly specified explanation, referencing an article that doesn't clarify the explanation. Say the evidence is caused by “derivatives” – just the one word is fine – and reference a long paper that doesn't explain how “derivatives” caused the evidence.

Ignore any existing disproof of any of your explanations. If an article disproved “earnings releases” as an explanation a decade and a half ago [14], don't mention that. If the reason “the shortness of the formal trading hours versus the 24 hour global trading cycle” doesn't work as an explanation has already been explained to you [15], don't mention that, either.

Don't check any obvious, easily checkable implications of any of your explanations. Don't check whether the positive overnight returns of the individual stocks in Figure 3 of Ref. [6] are concentrated around earnings announcements. If you check and they are not – and of course they are not – don't mention you checked.

Do not acknowledge any errors. If any objective, factual errors you have made are explicitly pointed out to you [11], do not acknowledge them. Do not correct your errors, either in what you have already written [10] or later in a subsequent article. If you are eventually forced to comment on the topic again, just restate your previous claims [13] without acknowledging any error in any of them.

Point to your regulators. You couldn't possibly have done anything wrong, because if you had, your regulators would have caught it. This is a great technique. It ingratiates yourself with your regulators. Spoken by your mouthpiece, it makes it your regulators' fault, not your mouthpiece's fault, if your fraud is eventually uncovered, giving your mouthpiece some cover. This technique relies on your victims and skeptics trusting your regulator, so your mouthpiece can't do anything to shake that trust. Your mouthpiece definitely can't directly ask your regulator to explain the evidence and publish their no comment (or otherwise unsatisfactory explanation) [10, 13, 16].

Repeatedly point to your regulators [10, 13]. This is such a great technique you can just keep doing it [38].

As long as your fraud is properly designed (Section III), these techniques for claiming there is nothing to see here will work beautifully. Nearly two years after Ref. [10] appeared in print, they are still working. Few people have recognized these brilliant techniques, those few have kept it to themselves, and nobody has raised a fuss.

This important point holds even if the cause of Figures 3-10 turns out to be innocuous. The errors in Ref. [10] are still errors, few people have noticed the errors, those that did don't have a good reason to believe Figures 3-10 are fine [5, 7], they kept the errors in Ref. [10] to themselves, and none of them raised a fuss.

All this is possible if and only if your fraud is properly designed.

III. WHEN YOU NEED TO

The best way to say there is nothing to see here is to never have to say it. The second best way is to only have to say it to people who don't want to see anything.

Most people focus on a fraud's mechanics, but good fraud design is all about incentives. If you get the incentives right, nobody will raise a fuss and everyone will accept whatever you say.

The mechanics of most frauds are simple. Wirecard pretended to have a couple billion euros in Philippine banks they didn't really have [17]. Bernie Madoff told his investors they had money they didn't really have [18]. The magic isn't the mechanics. The magic is the incentives. The magic is making people believe. The magic is making people want to believe.

When embarking on an activity as potentially rewarding as your own financial fraud, you want to learn from the best. The best financial fraud we know of is the Strategy. It might be the best financial fraud ever committed. The incentives the Strategy creates are so incredible you've probably never even heard of it.

The mechanics of the Strategy [1-3] are simple. Start with a large, market-neutral portfolio of stocks. Every day, when the market is illiquid (particularly before, at, and just after market open), expand your positions, buying more shares of the stocks you are already long and shorting more shares of the stocks you are already short. Every day, when the market is liquid (from later in the morning to market close), contract your positions, selling the shares of the stocks you bought earlier in the day and covering the additional shorts you added earlier in the day [39]. Trading when the market is illiquid moves prices more than trading when the market is liquid, so the net effect of your repeated daily round-trip trading is to move prices in your favor, creating mark-to-market gains on your existing portfolio. Your mark-to-market gain (in units of dollars) is proportional to the size of your existing portfolio. The cost of your repeated round-trip trading (in units of dollars) does not depend on the size of your existing portfolio. Your gains will therefore exceed your cost as long as your portfolio is sufficiently

large [1-3].

Wirecard just made up money that didn't exist. Madoff just made up money that didn't exist. The Strategy also makes up money that doesn't exist. Manipulating the market moves the "fair" market price, and mark-to-market accounting makes moving the "fair" market price in your favor just as good as making up money.

Making up money is great. Making up money doesn't produce disgruntled victims along and along. Victims are whiners. Victims make a fuss. You don't want people to make a fuss while your fraud is running. The Wirecard and Madoff frauds worked well (while they were working) because they didn't produce people who made a fuss. While your fraud is running, don't produce whiny losers. Make everybody you touch a winner.

By manipulating prices in a large market like the world's stock markets, the Strategy makes up *lots* of money. So much money. Tens of trillions of dollars. Tens of *trillions* of dollars. With a *t*. Multiples of \$10,000,000,000,000. Enough money to make everybody the Strategy touches a winner.

A firm (let's call it M) running the Strategy pays a prime broker, often a large bank, to finance M 's positions. M pays a commission for the use of a service, often provided by a large bank, that submits orders electronically to exchanges. M pays a fee to the exchange itself for each transaction. M 's daily buying high and selling low is a transfer of money from M to market makers and other professional market participants. In the United States, M pays a transaction fee that funds M 's regulator, the SEC [40]. M 's mark-to-market gains make money for M , M 's investors, and M 's employees. M 's market manipulation makes money for everyone who owns stocks. All this made up money increases the prices of other asset classes, too, like real estate, so M 's stock market manipulation also makes money for everyone who owns a home. The real estate industry loves steadily increasing house prices because of the many ways steadily increasing house prices make the real estate industry money. The financial industry loves steadily increasing stock prices because of the many ways steadily increasing stock prices make the financial industry money. While the Strategy is running, it doesn't produce disgruntled victims along and along. Everybody the Strategy touches is a winner.

The Strategy does produce victims along and along, of course – everybody the Strategy doesn't touch is, relatively speaking, a loser. As the Strategy runs and time passes, these losers, observing the winners, will get the sense something isn't right and will become disgruntled whiners. But their whining will be generic and unfocused and their anger will not be aimed at M or the Strategy because they won't even know about M or the Strategy because nobody will tell them.

If your fraud is small you may escape attention, but we recommend going big. Really big. Absurdly big. So absurdly big nobody would believe you even if you told them with a straight face exactly what you were doing. That way if anybody ever tries to tell people with

a straight face exactly what you are doing, they won't believe it.

Pair your fraud with a real business. Wirecard provided real payment services. Madoff provided real brokerage services. In addition to running the Strategy, *M* can run other strategies, placing bets in which he has an advantage because he knows he is systematically pushing up prices. Your legitimate business doesn't need to be profitable. It just needs to provide a story people can tell one another to explain your success.

Your regulators won't be a problem. They are stretched thin, poorly resourced, and will always be playing catch-up. If anyone complains about you to your regulators, your regulators will come to you and ask you for your explanation. Your regulators lack the authority and willingness to demand all the information they need, and they lack the time, manpower, and expertise to properly process that information even if they did. If you are sufficiently big and established, you will have plenty of money to pay lawyers and others to delay and obstruct your regulators' efforts. Aware of this, your regulators won't even try. We aren't dissing the dedicated public servants doing their best or suggesting we could do better. We're just pointing out that in the financial industry, the resources of the regulated grossly exceed the resources of the regulators, who have an impossible job.

Your regulators will play a crucial role in your fraud. Your regulators must be viewed by outsiders as reasonably competent, so those outsiders will view your regulators' inaction against you as a sign you can be trusted. But your regulators must be viewed by insiders as incompetent, so any insider who might blow the whistle on you won't because he understands your regulators won't do anything. And your regulators must in fact be incompetent, so if any insider actually does blow the whistle on you your regulators indeed won't actually do anything.

You can add language to your employment agreements to further discourage insiders from blowing the whistle on you. Language used by D.E. Shaw & Co. that recently became publicly available [19] may provide inspiration. Your regulators may eventually make you change your language [19], but it will take them a decade to do it [19]. In the meantime, every year that passes, your fraud grows bigger. In the meantime, every year your regulators don't catch you makes it harder for them to expose you. Regulators aren't going to voluntarily expose a big problem they missed for a decade. (That's just how people and institutions are – no disrespect to regulators *per se*.) Eventually, your regulators will put more effort into covering up their failure than trying to expose you. Eventually, your regulators' need to save face will make them your active protector. Befriend them. Help them out with stuff. When they visit your office, give them free snacks.

The Strategy does a similarly beautiful job of keeping journalists from spilling the beans. Any journalist is going to be net long, not net short, the stock market – if not directly by owning stocks, then indirectly by (partially)

owning a house, or indirectly by their choice of occupation – and he is going to think twice before calling out a potential problem if his very act of calling it out might suddenly reduce his own net worth. He could first reduce his net long position, but regulators are apt to cry foul at that, and if he is married, he is probably going to have to have a conversation with his spouse first, so that then becomes a whole thing. If the journalist nonetheless proceeds, his editor and everybody else who will eventually get pulled into the decision of whether to run the story will face the same issues. The higher up the chain, the greater the focus on the business consequences of running the story, all of which are bad. Exposing a problem in the stock market in a way that might cause prices to fall will hurt the journalist's company and every other media company he might try to jump to. The company will lose subscribers when its subscribers suddenly have less money. The company will lose advertising revenue when its advertisers suddenly have less money. A company funded by donations will lose donations when its donors suddenly have less money. A company with a billionaire owner that runs a story causing its owner to lose lots of money can expect personnel changes. Journalists are professionals, but they are also employees. A journalist may view his job as informing his readers, but the real job of any employee is to help his employer make money. Whenever such roles come into conflict, the role of the employee wins, and any employee who forgets that will be replaced with a different employee.

Nobody in the financial industry is going to speak out against his personal interest, so everyone in the financial industry a journalist might ask about the Strategy or evidence a firm is running it will do their best to downplay the possibility. The journalist, not wanting to find a problem and not wanting to burn the contacts representing his professional lifeline he has spent his career establishing, isn't going to push back.

M can rely on journalists and big media companies to serve their own (and *M*'s) interests in their role as gatekeepers by keeping striking evidence of *M*'s use of the Strategy from the public. *M* doesn't need to worry about anyone trying to make an end run around the gatekeepers using social media, either, because everybody wants the market to go up and because *M*'s use of the Strategy is so big it is *prima facie* absurd. As a backup, though, just in case, Figure 2 of Ref. [5] shows suspicious overnight and intraday returns in the stocks of those controlling your Facebook, Instagram, Apple News, and Twitter feeds and the results you get when you google or use Siri or Alexa. Any startup social media alternative that eventually wants to go public, be bought, or even just continue to get funding will desperately want the stock market to continue to climb and therefore won't be a problem, either.

If you are also running a legitimate business (as we strongly advise) and any reporter does come to you, invite him to interview other people at your business, particularly senior people heading business units sufficiently

removed from your fraud. To be safe, you yourself can be on vacation, or ill, or in a meeting when your appointed interview time rolls around [41]. Presented with the opportunity to interview senior members of your enviably successful firm, the reporter himself will orgasm, and the resulting article will have such a radiant postcoital glow you will link to it from your company’s web page for years.

Like others, most government officials have a direct personal financial interest in not calling out a problem if their calling it out might cause stock prices to fall. Unelected officials already have plenty of problems on their plate without adding another. Elected officials have nothing to gain and their office to lose if stock prices fall (for whatever reason). They can tell themselves they are hiding the problem from the public for the public’s own good, despite knowing full well that the public will eventually find out and the problem will just continue to grow until they do.

Academics have the same problematic personal financial incentives as others. Their professional incentives are a bit different, but not sufficiently different to jeopardize M . In academia, the first person gets all the credit, and nobody gets any credit for agreeing. The first academic to figure out what M is up to has an incentive to point out what M is up to. After that, no academic has any incentive to do anything. As time passes, any academic who probably should have done something but didn’t stands to lose face. As time passes, all academics but the first acquire significant professional and reputational incentives to keep others from hearing about the problem and their own inaction.

A thread with the SEC and others publicly available at Ref. [7] documents the extraordinary power of these incentives. The first few years of the thread (anonymized) were made available with Ref. [5]. From April 20, 2022 onward, the thread has been publicly maintained without anonymization, explicitly noting to recipients the public availability of the thread. The past year and a half saw the addition of over 150 individual emails at a rate of roughly two emails per week. Each email on the thread highlights a strikingly suspicious pattern of overnight and intraday returns in a different publicly traded stock. For the most part we have chosen stocks from the S&P 500 index in the United States, going roughly alphabetically by company name (and just recently reaching the companies whose names start with the letter P). Here and there we highlight other stocks that caught our eye. Almost all the words in the thread are ours. Excluding *pro forma* responses from regulators and a few automatic undeliverable and out-of-office messages, we received exactly one reply. One one-line reply.

Each email is addressed primarily to the SEC’s chair or commissioners, and secondarily to one or more of the sorts of people you might imagine would alert the public to suspicious return patterns with no apparent innocuous explanation in the world’s stock markets – mostly financial regulators, other public officials, finance profession-

als, economists, and journalists. The first email in each quarter is addressed to regulators of public equity markets around the world. All plots use data publicly available from Yahoo! Finance [8] and range from the start of 1990 (or the first date for which good data are available) to the end of the most recent quarter [42]. The thread is about these strikingly suspicious return patterns in the world’s stock markets, the apparent absence of any plausible innocuous explanation for them, and the fact that the public doesn’t know about them because nobody has told them. Each of these three topics is objectively factual, easily verifiable, and undisputed. The thread is very deliberately not at all about M , the Strategy, and the explanation we have suggested for these strikingly suspicious plots [1–3].

Many of the thread’s recipients were previously aware of these strikingly suspicious return patterns, either because they themselves were among the hundreds of people we privately, individually, more diplomatically reached out to over the preceding five years or because they would have heard of this issue from others. They are people who would not be shy about pointing out a compelling alternative explanation or relevant evidence we had overlooked – to their usual audience through their usual communication channel, if not on the thread itself. They are people who could potentially alert the public.

The primary goal of the thread was to encourage someone with a megaphone larger than ours to alert the public. The secondary goal, failing the first, was to document the failure.

In documenting the failure, our point is not to highlight any specific individual’s inaction. Perhaps he faced some other pressing deadline that took all his time, for months. Perhaps he is still trying to learn how to make a plot himself. Whatever. The point of the thread is not any specific individual’s inaction. The point is the pattern.

The pattern is hundreds of people who all understand the difference between positive numbers and negative numbers, who all understand the attached plots of overnight and intraday returns should not look like they do, who all have colleagues who can reproduce the numbers and plots themselves, who all understand they do not have a compelling innocuous explanation for the plots or a good reason they are fine, who all have colleagues who can review the literature and confirm the apparent absence of a compelling innocuous explanation for the plots and a good reason they are fine, and who, to a person, have chosen to not alert the public to these objective, easily verifiable, undisputed, disquieting facts.

This thread [7] is great news for you. It is long, but we encourage you to read it in full to appreciate the level of egregious behavior you can get away with without anyone making a fuss as long as you have properly designed your fraud’s incentives.

The incentives created by the Strategy are so good they even provide M with an endgame when M ’s use of the Strategy is inevitably eventually exposed and all the made up money disappears. By being big and having

taken plenty of money off the table along and along, M will have plenty of money for lawyers and plenty of money left even after paying big fines. By being big, M 's use of the Strategy has produced many people the public will blame. A few heads will need to roll, but most of the perps will get to keep their own and enough money to make the whole exercise very worthwhile.

Similarly, as long as you design your fraud with our guidance and ensure you are not the sole face of it, your risk is not high, even in the endgame. In criminal court, prosecutors have the burden of proving knowledge and intent, the burden of telling a jury a simple story, and little incentive to try cases they aren't sure they will win. In civil court, you just need to keep judgments less than the money you made. In the court of public opinion, the story will be simplified to emphasize a villain or two – nobody casually following along at home is going to bother keeping track of more – and with some combination of maneuvering and luck there's a decent chance your face will not be among the few the public learns to hate.

To summarize, you can produce global, decades-long, blatant evidence of your malfeasance and get away with it as long as nobody has an incentive to make a fuss. This important point holds even if the cause of Figures 3-10 turns out to be innocuous. Many people in a position to alert the public know about Figures 3-10 [7], don't have a compelling innocuous explanation for Figures 3-10 or a good reason to believe Figures 3-10 are fine [5, 7], and kept these disturbing facts to themselves. None of them raised a fuss.

IV. GO GET 'EM

You now know what you need to launch your own financial fraud. Section I provided confidence-building examples of what you can get away with. Section II provided brilliant techniques for how to say there is nothing to see here. Section III described the sort of incentive structure you want to create with your own fraud and the effect these incentives will have on people.

Importantly, the lessons of each of these three sections hold even if the cause of Figures 3-10 turns out to be innocuous. Even if Figures 3-10 turn out to be fine, they show the sort of evidence you can produce and get away with without anyone raising a fuss despite nobody having a good reason to believe they are fine. Even if Figures 3-10 are fine, the *Financial Times* has provided a great example of how to say there is nothing to see here, receiving no pushback even though every single reason they provided is clearly objectively wrong. Even if Figures 3-10 are fine, our thread with the SEC [7] shows the world's financial regulators, journalists, economists, finance professionals, and others with no good reason to believe Figures 3-10 are fine not bringing these strikingly suspicious return patterns to your attention because of the powerful incentives we describe.

Your fraud will be great. Go get 'em.

If you are still hesitant, show a few of the plots in Figures 3-10 to your friends and practice saying the words.

There is nothing to see here.

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- [19] *SEC Charges D. E. Shaw with Violating Whistleblower Protection Rule*, September 29, 2023, press release available at <https://www.sec.gov/news/press-release/2023-213>, SEC Order available at <https://www.sec.gov/files/litigation/admin/2023/34-98641.pdf>.
- [20] Given our choice of plot format in Figure 1(c), it is all but inevitable that at some point a figure will be lifted from this article without its caption and someone seeing it will complain you can't take the log of a negative number. When that happens, please gently point the complainer to Figure 1 and convey our apologies for the confusion.
- [21] The curves in Figure 3 start on 1990-01-01 or the first day for which good data from Yahoo! Finance are available: S&P 500 SPDR ETF (1993-01-29), iShares Russell 2000 ETF (2000-05-26), S&P 400 Mid-Cap SPDR ETF (1995-05-04), iShares TSX 60 ETF (1999-10-04), HSBC FTSE 100 ETF (2009-08-25), CAC 40 (1990-03-01), IBEX 35 (1999-02-23), DAX (1993-12-14), AEX (1992-10-12), BEL 20 (1992-06-10), DNB OBX ETF (2009-05-08), iShares FTSE MIB ETF (2009-01-02 to 2023-03-24), TA-125 (2007-01-08), NIFTY 50 (2007-09-17), S&P BSE SENSEX (1997-07-01), SPDR S&P/ASX 200 Fund (2009-01-02), SET Index (1996-12-11), KOSPI Composite Index (1996-12-11), TSEC Weighted Index (1997-07-02), SPDR Straits Times Index ETF (2009-01-02), and SSE Composite Index (1997-07-02).
- [22] Figure 2 of Ref. [3] showed the iShares Core FTSE 100 ETF (symbol ISFU.L on Yahoo! Finance) starting in 2001, but those data are no longer available, so Figure 3 shows the HSBC FTSE 100 ETF (HUKX.L). At the time of this writing, the iShares FTSE MIB ETF (symbol IMIB.MI on Yahoo! Finance) has bad prices for a few weeks starting in late March 2023, so the plot for Italy's FTSE MIB in Figure 3 ends on 2023-03-24. The FT in FTSE is the *Financial Times*, so the quality of data for indices bearing the FT's name is pretty much what you would expect from Section II.
- [23] A ".PA" suffix is implied on all symbols except MT.AS. All 40 constituents of France's CAC 40 index are shown with a linear vertical scale on July 4, 2023 in Ref. [7]. France's CAC 40 index constituents are available from <https://live.euronext.com>.
- [24] France's Autorité des marchés financiers (AMF), <https://www.amf-france.org>.
- [25] A ".DE" suffix is implied on all symbols. All 40 constituents of Germany's DAX index are shown with a linear vertical scale on October 1, 2023 in Ref. [7]. Germany's DAX index constituents are available from <https://www.boerse-frankfurt.de>.
- [26] Germany's Bundesanstalt für Finanzdienstleistungsaufsicht (BaFin), <https://www.bafin.de>.
- [27] A ".L" suffix is implied on all symbols. The UK's FTSE 100 index constituents are available from <https://www.londonstockexchange.com>.
- [28] The UK's Financial Conduct Authority (FCA), <https://www.fca.org.uk>.
- [29] A ".NS" suffix is implied on all symbols. India's Nifty 50 index constituents are available from <https://www.nseindia.com>. Plots of overnight and intraday returns to the same stocks trading on the Bombay Stock Exchange are qualitatively similar.
- [30] A ".T" suffix is implied on all symbols. Japan's Nikkei 225 index constituents are available from <https://indexes.nikkei.co.jp>.
- [31] Japan's Financial Services Agency (FSA), <https://www.fsa.go.jp>.
- [32] A ".SS" suffix is implied on all symbols. China's SSE 50 index constituents are available from <https://www.csindex.com.cn>. Figure 4 of Ref. [6] shows all 50 constituents of the SSE 50 with a linear vertical scale.
- [33] *Skandal! Bringing Down Wirecard* (2022), directed by J. Erskine, URL <https://www.netflix.com/title/81404807>.
- [34] *Inside Wirecard*, *Financial Times*, URL <https://www.ft.com/wirecard>.
- [35] This article would have addressed compelling alternative explanations for Figures 3-10, good reasons Figures 3-10 are fine, and/or evidence disfavoring our explanation [1-3] for Figures 3-10, but in the nearly two years since our last article [6], no compelling alternative explanations, good reasons, or such evidence has appeared. We do not pick on the *Financial Times* because they are an amusing distraction from serious points others have raised. We pick on the *FT* because in the nearly two years since our last article, they are the only ones who provided new material for us to work with. Our previous articles [1-6] have already addressed everything else.
- [36] We are not accusing the *FT* of being the mouthpiece of a firm committing financial fraud. We are saying when you commit your financial fraud, you want a mouthpiece who will do what the *FT* has done.
- [37] Use upbeat adjectives. Call suspicious return patterns in the world's stock markets suggesting global, decades-long market manipulation "curious" [10], "delightfully weird" [10], "wonderfully weird" [13], and "fascinating" [13].
- [38] The *FT*'s greatest recent investigative success – worthy of a Netflix documentary starring real people at the *Financial Times* [33] – is its exposure of Wirecard's fraud [34]. The *FT*'s investigation of Wirecard succeeded despite active interference from Germany's financial regulators. Germany's financial regulators' failure to catch Wirecard's fraud is not lost on anyone at the *Financial Times*, making the *FT*'s "if this were a problem, a regulator would have addressed it" claim [10, 13] even less believable and more seemingly disingenuous than it would be coming from someone else.
- [39] Your actual trading can be complicated in detail and still retain the essential "expand your portfolio when the market is illiquid, contract your portfolio when the market is liquid" pattern that is the essence of the Strategy.
- [40] Congress sets the SEC's budget. The SEC sets the value of the transaction fee so the total expected fees collected match the SEC's budget.
- [41] The resulting article will not contain a quote from you, but only insiders close to the story who know you are pithy and eminently quotable will recognize this telling omission, and none of them will say anything.
- [42] In a few timely cases (like an impending bankruptcy), the plot's end date is closer to the date of the email than the end of the preceding quarter.

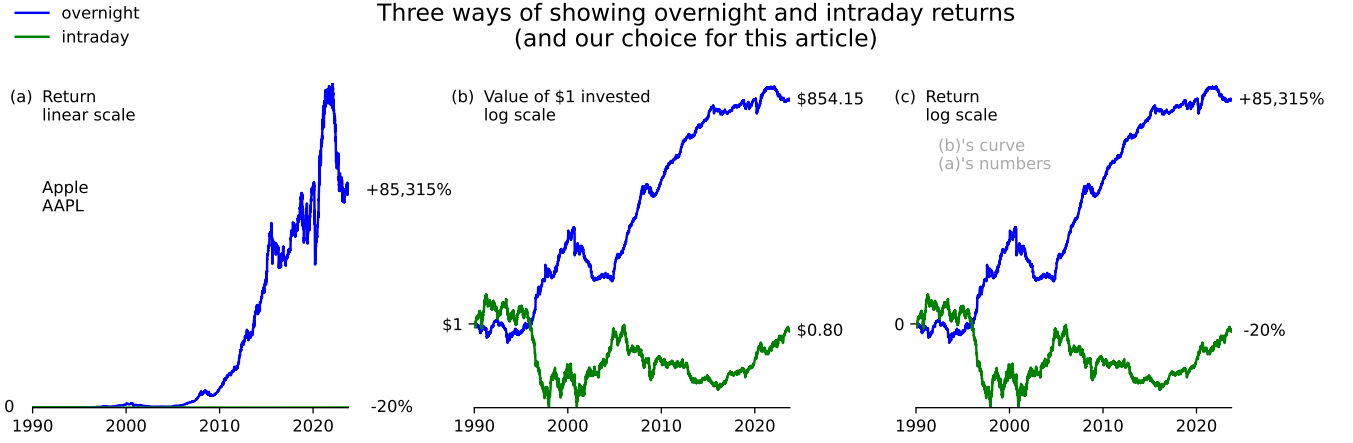


FIG. 1: Three ways of showing overnight (blue) and intraday (green) returns, and our choice for this article. The cumulative intraday return to Apple’s stock (from market open to market close) from January 1, 1990 to September 30, 2023 is -20% . The cumulative overnight return (from market close to the next day’s market open) is $+85,315\%$. In (a), the cumulative overnight and intraday returns are shown with a linear vertical scale. The linear vertical scale is intuitive, but the green curve gets buried. In (b), the same data are recast as the value of \$1 invested in Apple’s stock, getting only overnight or intraday returns, and displayed with a logarithmic vertical scale. Starting with \$1, a return of $+85,315\%$ leaves you with \$854.15, and a return of -20% leaves you with \$0.80. Panel (c) shows the curve from (b) and the numbers from (a). Panel (c) is potentially confusing and misleading (and does not involve the log of -20%) [20], but it nicely shows the informative log curve of (b) and the easily understandable returns in (a). We use the format of (c) for all plots in this article.

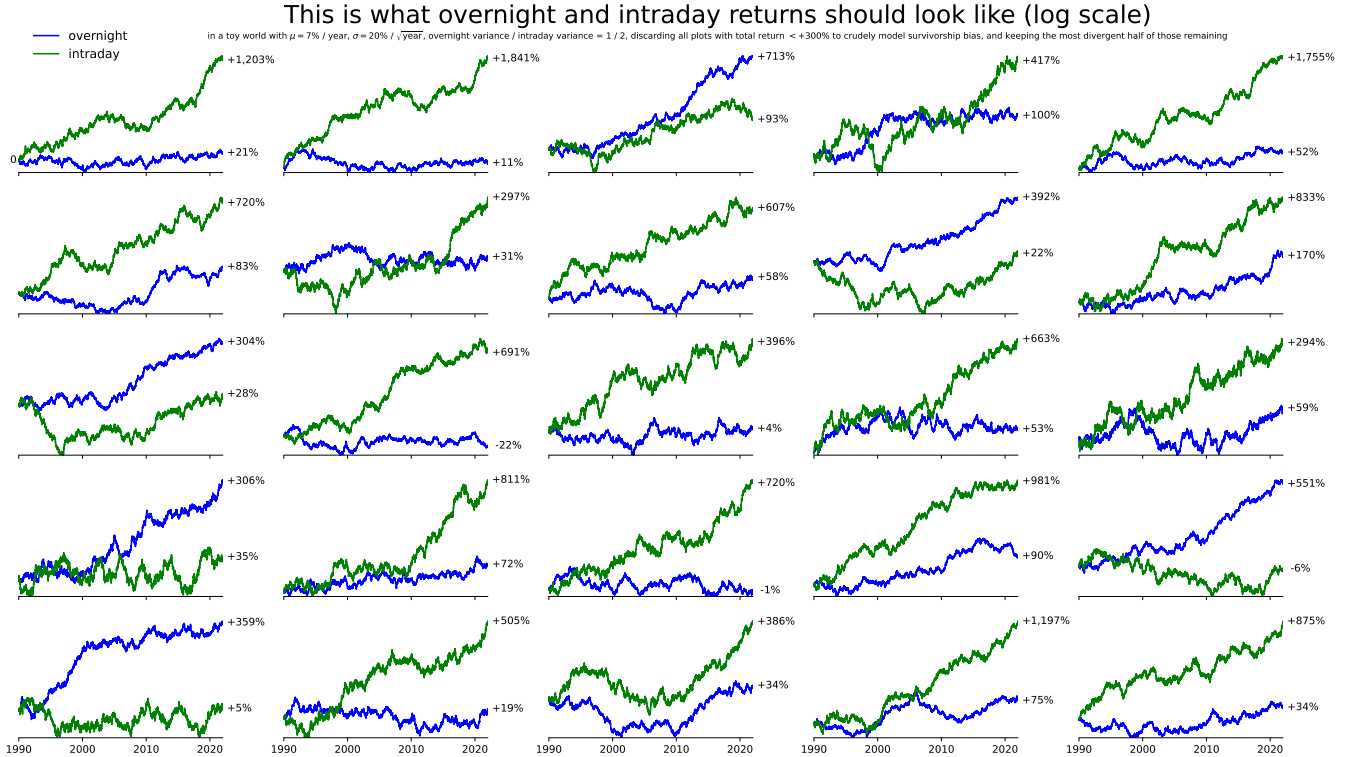


FIG. 2: Examples (generated with random numbers) of what plots of overnight and intraday returns should look like, in the format of Figure 1(c). To (at least partially) mimic the selection bias of subsequent figures, which show particularly problematic-looking stocks selected from a larger group, this figure shows the 25 most problematic-looking plots from the 50 plots in Figure 1 of Ref. [6]. Details are provided in Ref. [6]. The code used to produce this figure is publicly available at Ref. [9]. If everything were copacetic, subsequent figures would look more or less like this one.

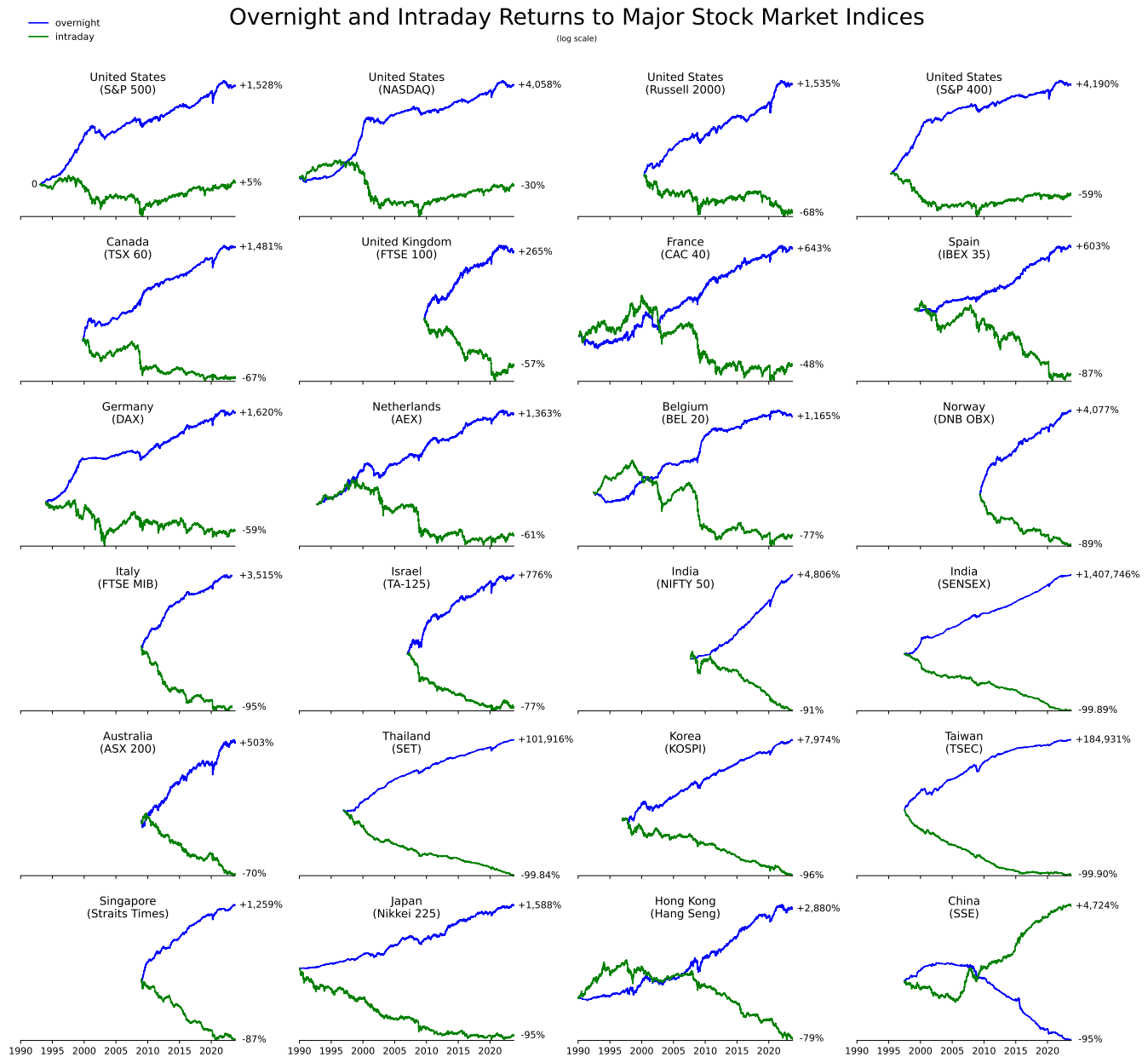


FIG. 3: Cumulative overnight (blue) and intraday (green) returns to twenty-four major stock market indices, in the format of Figure 1(c). Each plot shows data from January 1, 1990 (or the first date for which good data are available [21, 22]) to September 30, 2023. If you had invested ¥1 in Japan's Nikkei 225 at the start of 1990 and had gotten only overnight returns, by September 30, 2023 you would have made ¥15.88, for a cumulative return of +1,588%. If you had gotten only intraday returns, you would have lost ¥0.95, suffering a cumulative return of -95%. Data and code are publicly available [8, 9]. Figure 2 of Ref. [6] shows a version of this figure with a linear vertical scale, in the format of Figure 1(a). Figure 1 of Ref. [5] shows a version of this figure in the format of Figure 1(b). None of the plots in this figure look like they should (cf. Figure 2). If you live in one of the countries shown, you can ask your country's financial regulator whose trading caused this suspicious return pattern in your country's index. As a general policy, if you see something suspicious in one of your investments, you should ask for an explanation. If you don't get an explanation that makes sense, you should take your money elsewhere. The burden of proof is on your regulator to convince you the apparent problem is okay, not on you to convince them it isn't.

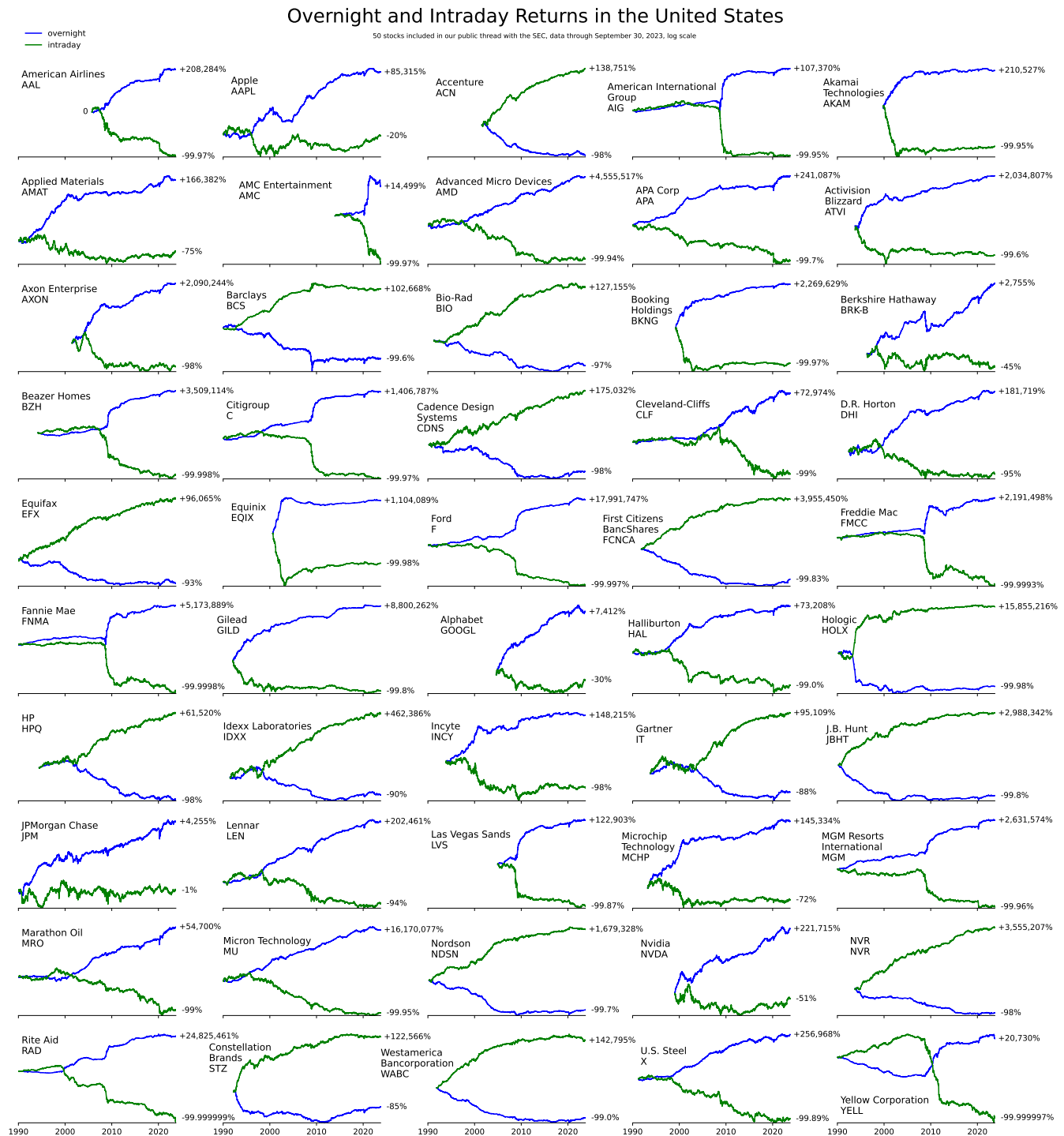


FIG. 4: Cumulative overnight and intraday returns to 50 stocks included in a public email thread with the United States Securities and Exchange Commission and others [7], shown in the format of Figure 1(c). As in all subsequent figures, data and code are publicly available [8, 9], and the data shown span the start of 1990 (or the first date for which good data are available) to September 30, 2023. None of the plots in this figure look like they should (cf. Figure 2). You can directly ask the SEC whose trading caused them. Several SEC email addresses are provided in Ref. [7]. Other points of contact are provided on the SEC’s webpage (<https://www.sec.gov>). As a general policy, if you see something suspicious in one of your investments that doesn’t make sense, you should take your hard-earned money elsewhere. You can explicitly ask for an explanation first, if you wish, but you don’t need to, and if a good explanation isn’t quickly forthcoming, you probably don’t want to hang around.

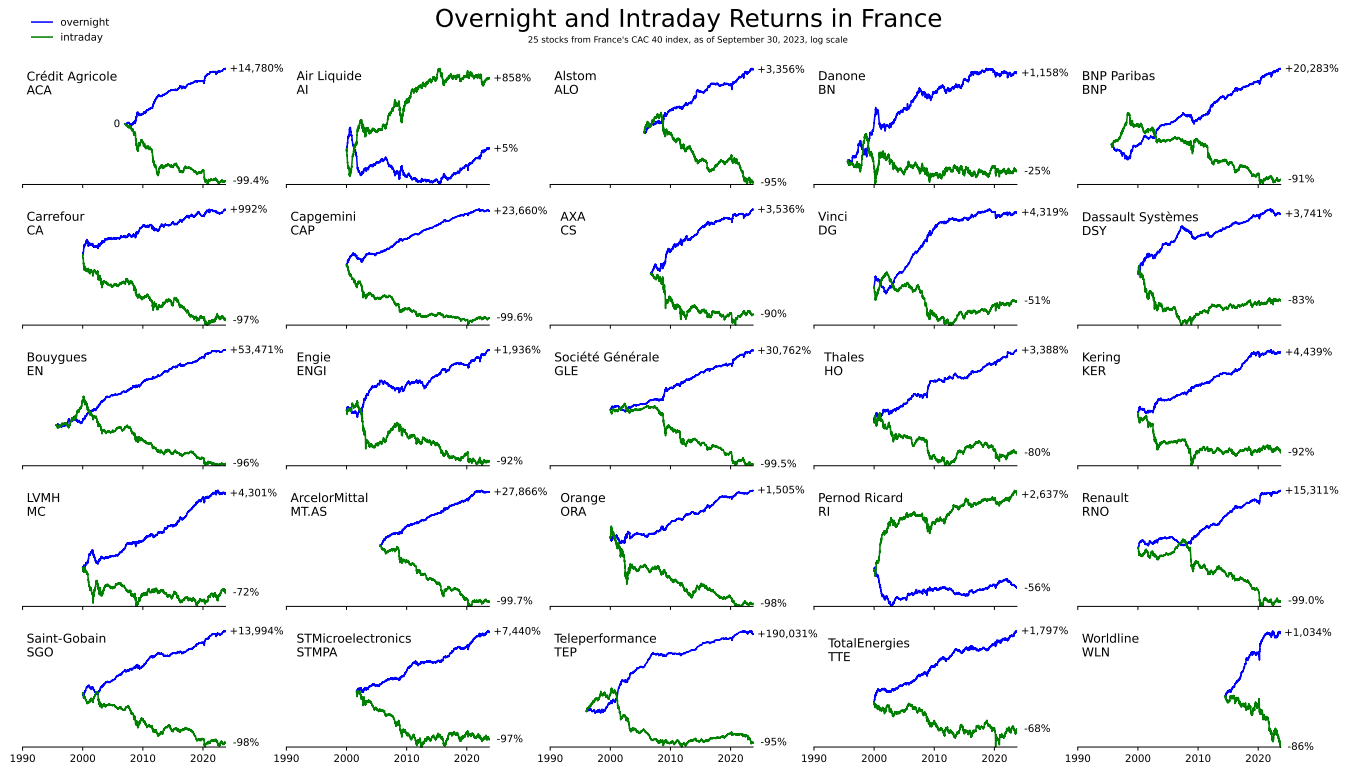


FIG. 5: Overnight and intraday returns to 25 stocks in France's CAC 40 index [23], in the format of Figure 1(c). If you are French and care about your country, you should ask France's financial regulator (directiondelacomunication@amf-france.org [24]) whose trading caused these riotous return patterns in the stocks of many of France's largest companies.

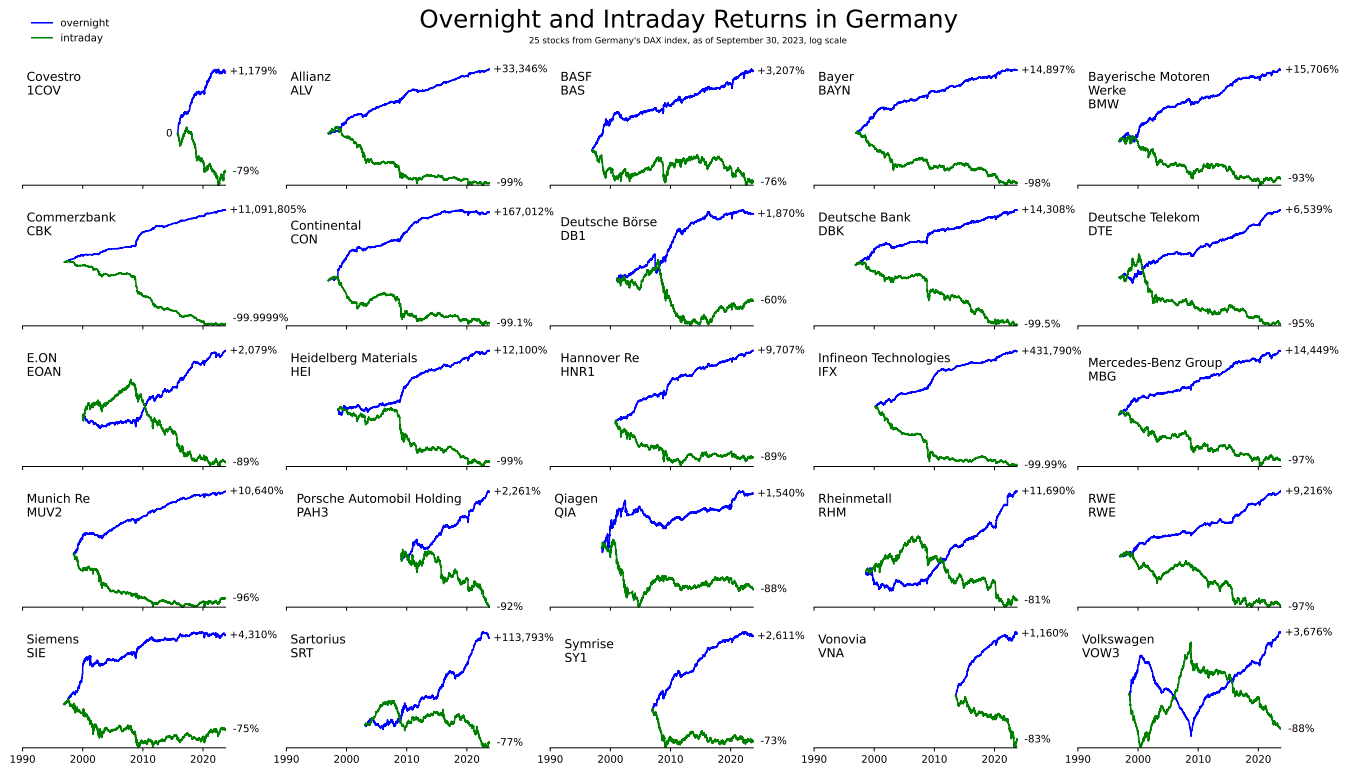


FIG. 6: Overnight and intraday returns to 25 stocks in Germany's DAX index [25], in the format of Figure 1(c). If you invest in Germany and care about your money, you should ask Germany's financial regulator BaFin (poststelle@bafin.de [26]) whose trading caused these scandalous plots.

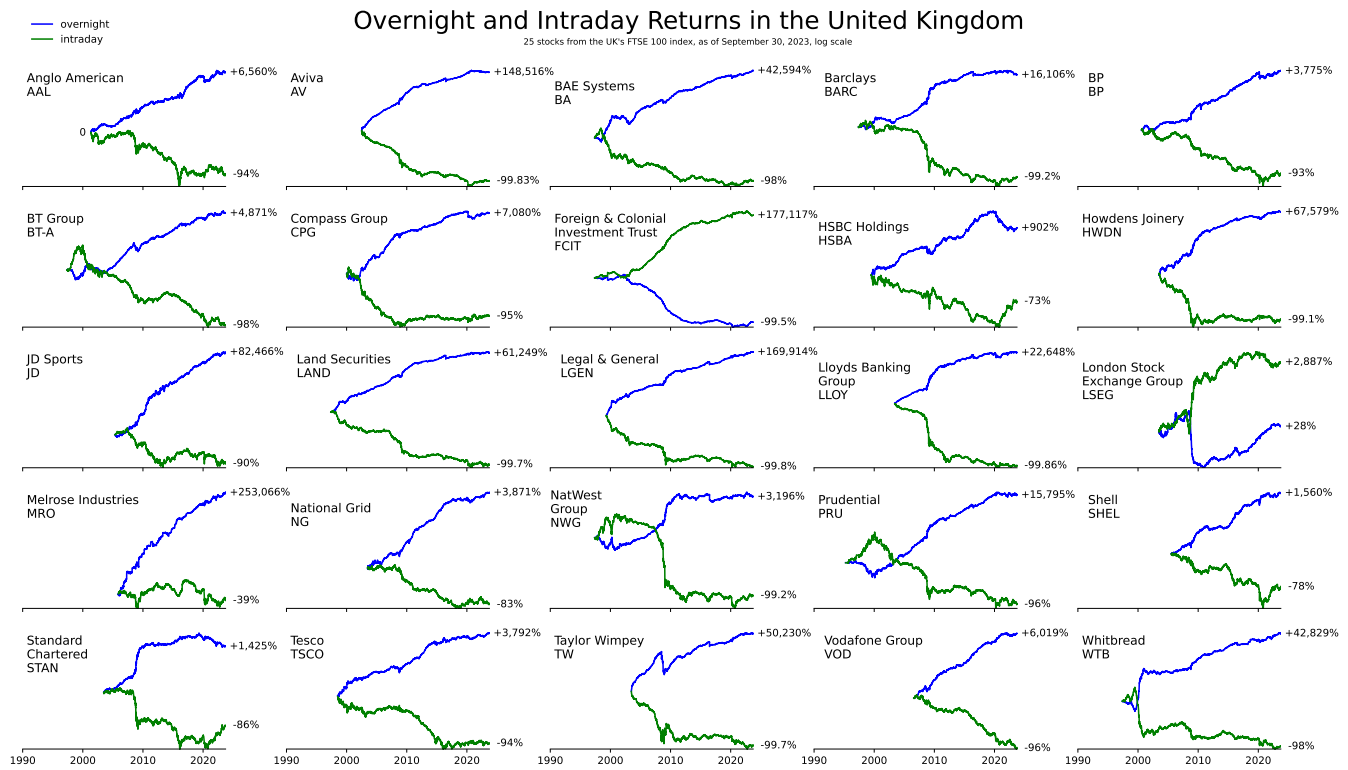


FIG. 7: Overnight and intraday returns to 25 stocks in the UK's FTSE 100 index [27], in the format of Figure 1(c). Ask the UK's regulator (consumer.queries@fca.org.uk [28]) whose trading caused these striking plots. If they can't give you a definitive answer, you should feel confident launching your fraud in the United Kingdom.

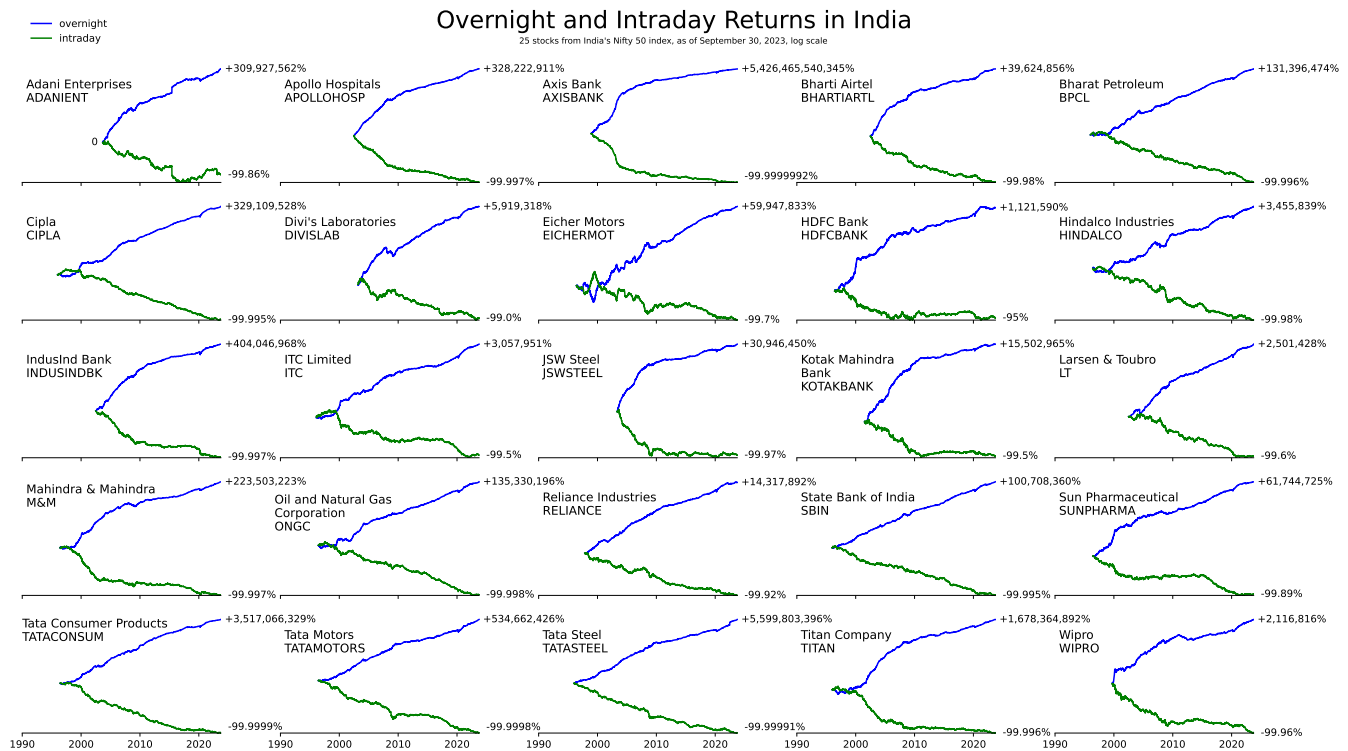


FIG. 8: Overnight and intraday returns to 25 stocks in India's NIFTY 50 index [29], in the format of Figure 1(c). India seems like a wonderful place to commit financial fraud.

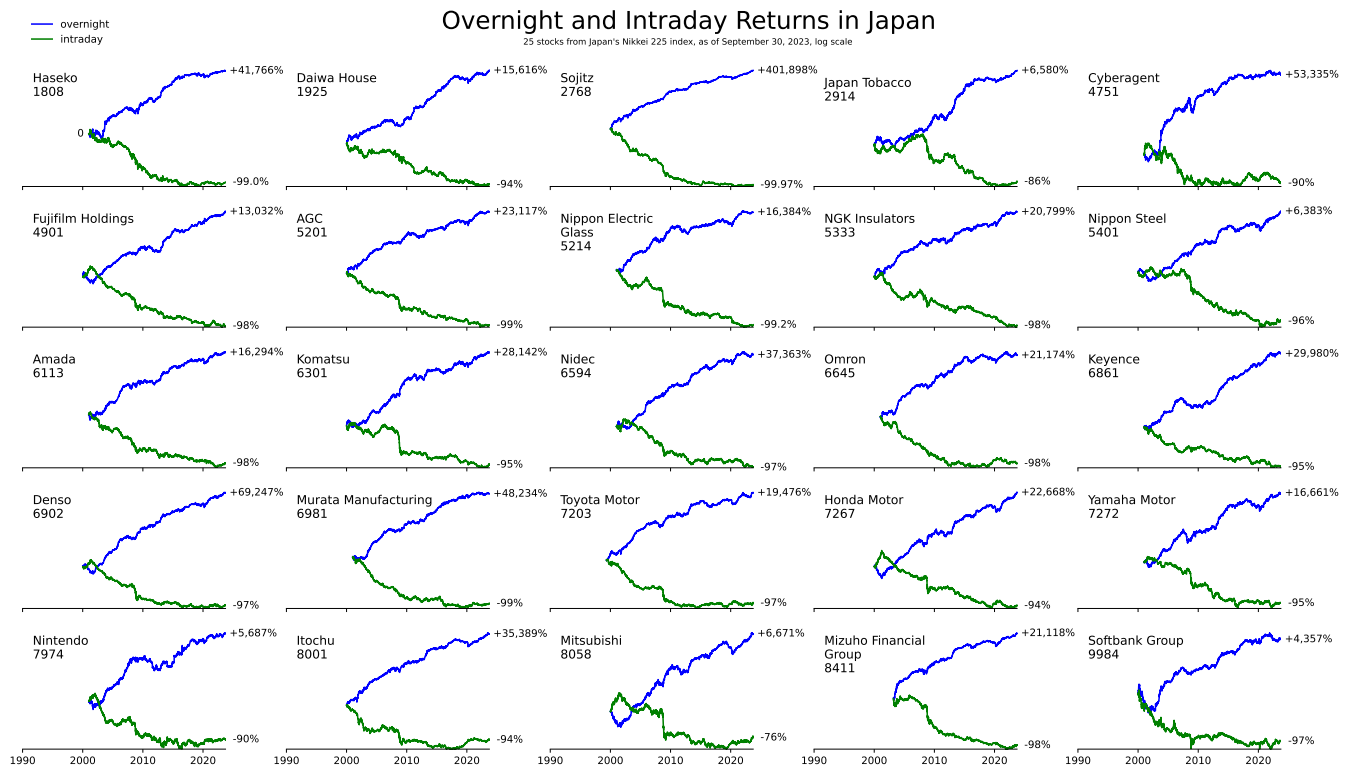


FIG. 9: Overnight and intraday returns to 25 stocks in Japan's Nikkei 225 index [30], in the format of Figure 1(c). Ask Japan's regulator (eququestion@fsa.go.jp [31]) whose trading caused these plots. If they can't answer, just think about all the fraud you'll be able to get away with in Japan.

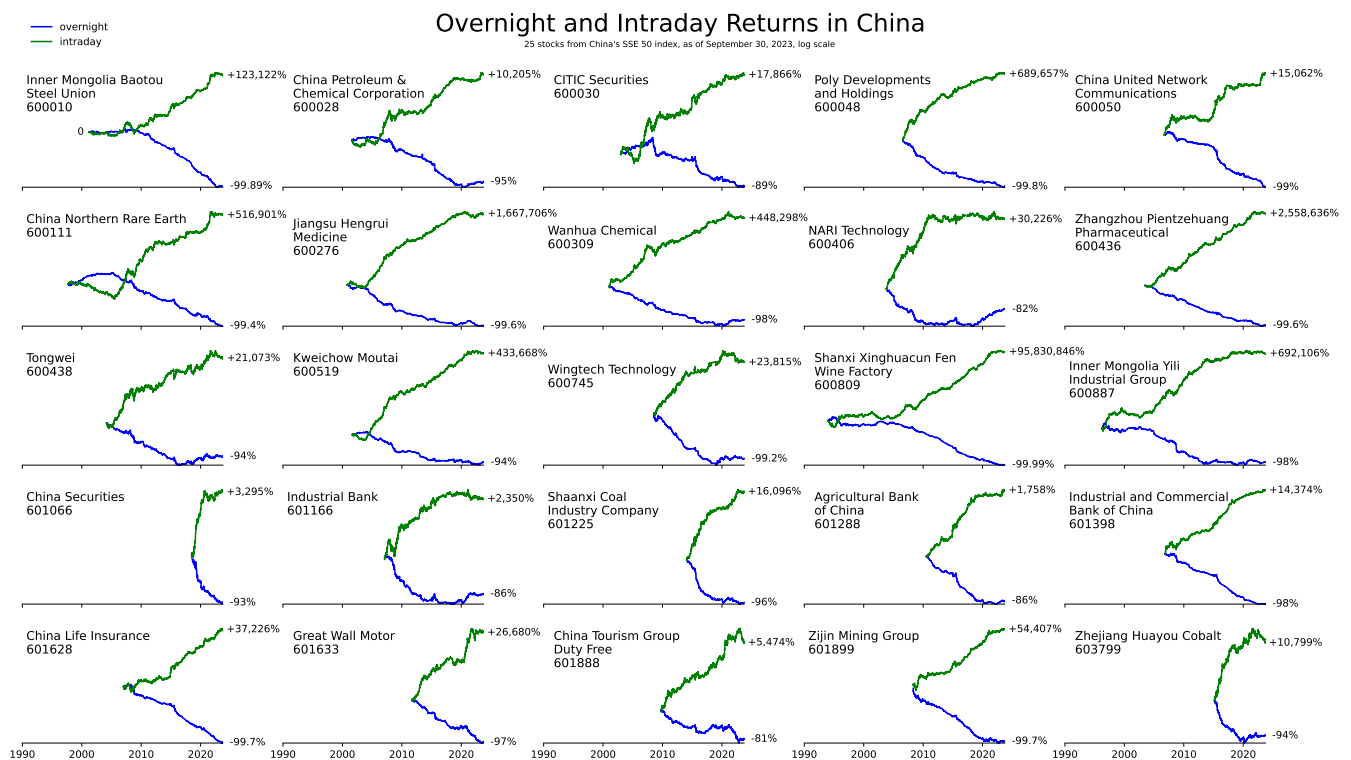


FIG. 10: Overnight and intraday returns to 25 stocks in China's SSE 50 index [32], in the format of Figure 1(c). If you are in China and don't have any family or friends who depend on you, consider loudly and publicly asking the Chinese government whose trading caused these plots and why it is taking them so long to figure it out.